SECOND YEAR LABORATORY

DETERMINATION OF ELASTIC CONSTANTS OF GLASS BY CORNU'S METHOD

1 Aims

To understand the meaning of Young's Modulus and Poisson's Ratio for an isotropic elastic solid and how to obtain values for these quantities by acquiring and analysing images of thin film interference patterns.

2 Objectives

1. Obtain values for Young's Modulus and Poisson's Ratio of a glass bar.
2. Utilise image acquisition and analysis techniques to measure the distance between fringes in optical interference patterns and use these to find values for curvature of the bar.

3 Introduction

Understanding how materials respond to applied force is of fundamental importance in many engineering and scientific applications. There are several different elastic moduli that characterise materials and the three primary ones are Young's, bulk and shear moduli which were covered in the first year Properties of Matter course (PHYS10352). Each elastic modulus has the general form of applied stress / resultant strain. This experiment uses an optical technique developed by Cornu in the 19th century that utilises thin film interference to measure the curvature of a glass bar when subject to a variable bending load.

4 Experimental setup

A schematic of the setup used for this experiment is shown in Figure 1. The setup consists of a glass bar that is supported on two knife edges that are symmetrically located on either side of the centre of the bar. A mass M is hung at a distance L from the knife edge at each side (as shown in Fig. 1). Both M and L can be varied across a range of values. This bending moment is how strain is applied to the glass bar. The resulting stress in the glass bar produces curvature along both the length (longitudinal curvature) and width (transverse curvature) of the bar.

The two components of curvature can be measured using an optical interference technique. A glass reference block with optically flat surfaces is placed at the centre of the top surface of the bar. Either a mercury or sodium discharge lamp is placed vertically above the bar with a yellow or green filter used to isolate the appropriate spectral line (the mean wavelength of the sodium D lines is 589 nm and the mercury green line is 546 nm). Some of the light is reflected from the bottom surface of the glass reference block and some of the light is reflected from the top surface of the glass bar (as shown in Figure 2). These two components of reflected light will interfere depending on their relative path difference. The path difference varies with position due to the curved surface of the strained glass bar.

The interference pattern is imaged by suspending a delicate semi-silvered glass sheet above the bar at 45° to the direction travelled by the light. This then reflects some of the light towards a camera connected to a computer that is used to record an image of the interference pattern.

Over time the large number of bending cycles that each glass bar experiences have likely caused them to gain some residual curvature when there is no applied stress (i.e. curvature when no masses are hung from the end of the bar).

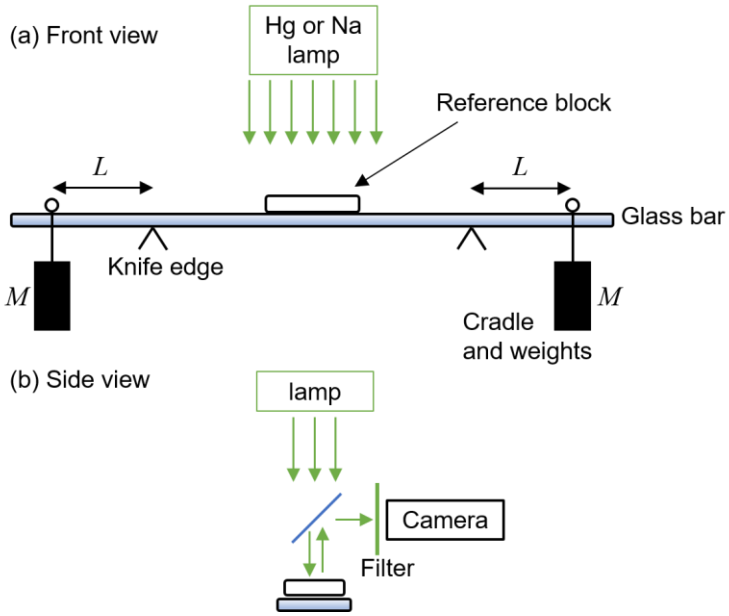


Figure. 1: Schematic setup of the experiment when viewed from (a) the front and (b) the side.

5 Theory

These are brief notes that outline that outline the theory of the experiment - for details see [1] and [2], or standard textbooks in the areas of physics concerned (e.g. optics and elasticity).

5.1 Thin Film Interference Fringes

The condition for destructive interference of light of wavelength λ is $2t = n\lambda$ where n is an integer and t is the thickness of the film, i.e., the distance between the two surfaces involved (shown in Figure 2), so $2t$ is the total path difference.

Question: Why does a path difference with an integer number of wavelengths produce destructive interference rather than constructive interference? What happens to light reflected at the glass-air surface compared to the air-glass surface (recalling that the refractive index for glass is greater than that of air)?

5.2 Curvature of the bar

Figure 2 shows the two surfaces that reflect light to produce the interference pattern.

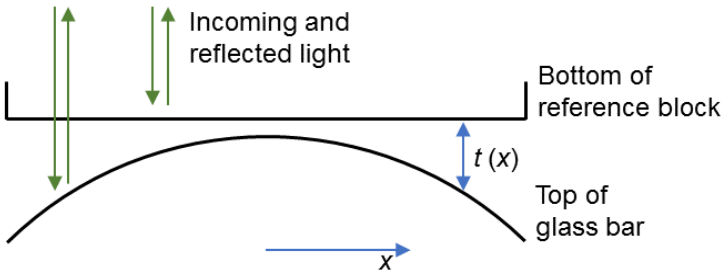


Figure 2: Illustration of the two surfaces that reflect light to produce the interference pattern. This is an example of thin-film interference, although the thickness of the film is spatially dependent in this case.

Using simple geometrical considerations show that

$$t(x) = t_0 + \frac{1}{2}\kappa x^2 \quad (1)$$

where t is the thickness of the air film between the surfaces, x is the distance from the centre (the point of minimum thickness), κ is the longitudinal curvature of the bar and t_0 is a constant (equal to the thickness of the airgap at $x = 0$). Note that curvature has dimensions of inverse length which means that the radius of curvature, $R = 1/\kappa$.

If x_n is the position of the nth dark fringe (due to destructive interference), you should also show that the following relationship holds:

$$\frac{n\lambda}{2} = t_0 + \frac{1}{2}\kappa x_n^2. \quad (2)$$

This exhibits a linear relationship between n and x_n^2 from which a value of κ can be deduced. (Note: it is easier to measure $2x_n$, the distance between a fringe and the corresponding fringe on the other side of the pattern - see Figure 4.)

5.3 Bending in a Bar

From [1], Section 56, the curvature κ caused by a bending moment G is

$$\kappa = \frac{G}{EI}, \quad (3)$$

where E is Young's Modulus and I is the moment of area of the cross-section of the bar about the neutral line (a line perpendicular to the bending plane through the centre of gravity of the cross-section).

From [1], Section 62, for a bar of rectangular section, as illustrated,

$$I = \frac{4}{3}ab^3, \quad (4)$$

where $2a$ is the width and $2b$ is thickness of the bar.

The bending moment in the central portion of the bar for the arrangement illustrated in Figure 1 is

$$G = MgL, \quad (5)$$

where M is the mass at each end of the bar and L is the distance between the knife-edge and the point of application of the weight. (This formula assumes the masses and knife-edges are positioned symmetrically with respect to the centre of the bar.)

5.4 Poisson's Ratio

Poisson's Ratio is defined as

$$\sigma = \frac{Fractional\ decrease\ in\ width}{Fractional\ increase\ in\ length} \quad (6)$$

for a specimen suffering uniform tensile stress. By considering the glass bar as a compound of thin longitudinal fibres it is possible to see that when the bar is carrying a bending moment causing a downward curvature κ of the longitudinal section, the transverse section obtains an **upward** curvature κ' such that

$$\sigma = \frac{\kappa'}{\kappa}. \quad (7)$$

Question: Check that you understand why the bar curves in this way. Can you explain this pictorially to someone else (e.g. your demonstrator)?

6 Obtaining images

Set up the apparatus as shown in Fig. 1 with moderate masses (1 kg is the recommended starting mass) suspended from each end of the bar. Be careful when hanging the masses on the bar to place the masses on each end at the same time as placing weights on one end of the bar only can cause the bar to quickly tip over. **The semi-silvered glass sheet is very delicate** and can easily be broken either by tipping the bar or by adequately not clamping the sheet in position – please take care.

You should make sure that the lower surface of the reference block and the corresponding upper surface of the glass block are clean. Lens cloths are available from the lab technician for cleaning these surfaces.

The suggested experimental arrangement allows you to set up an approximately constant bending moment over the central region of the glass bar. The resulting curvature can be measured by the optical interference method outlined in the previous Section.

The imaging is achieved using a Canon EOS-4000D digital camera connected to a computer. It is recommended that the camera lens is set to the highest zoom setting of 55 mm. Switch the camera to manual (M) mode to turn it on and remove the lens cap. Open the Canon EOS Utility software package on the computer. Select "Remote Shooting" and then click "Live View shoot" to allow you to see what the camera is seeing. A labelled screenshot of the software is shown in Fig. 3.

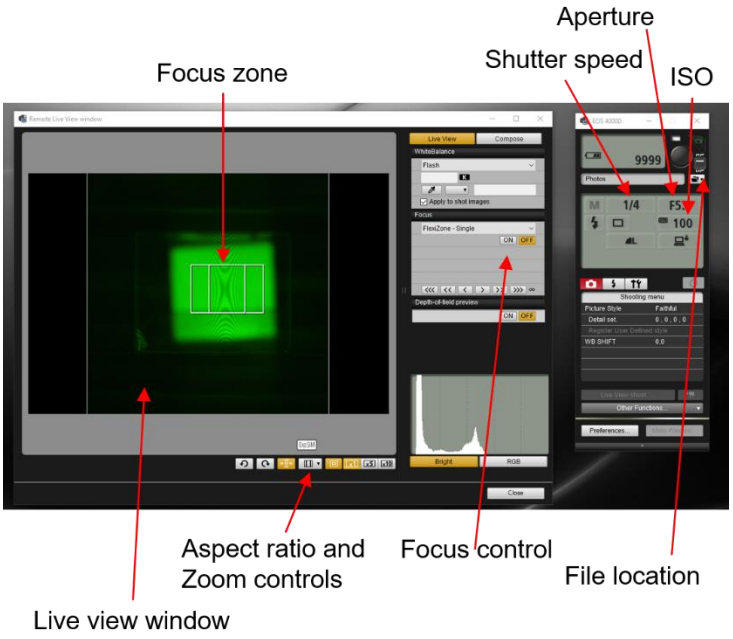


Figure 3: Labelled screenshot of Canon EOS Utility software.

The ISO should be set to a low value to reduce noise in the image (e.g. 100). The aperture needs to be wide to allow as much light as possible into the camera (so the F-number should be set to a low value, e.g. F5.6). A long shutter speed will be required such as 1/4 s (but there could be some variation in this value depending on the setup). The autofocus setting can be used. Click and drag the white rectangle in the Live View window to the region of interest (e.g centre of fringe pattern) and with the focus item set to "FlexiZone – Single" click "ON" to focus the camera. It is suggested to set the aspect ratio to 1:1 and the image quality to "L". If desired it is possible to set the picture style to "Monochrome" to obtain greyscale images (you should consider whether there are any advantages to doing this).

It is possible to choose the folder where you wish for image files to be saved. It is recommended that you change the filenames to something meaningful and document the details associated with each file in your lab notes. Before leaving the lab make sure that all image files are saved to a folder on your personal Onedrive (or equivalent cloud storage).

With the initial starting weights in place, adjust the position of the camera (and if necessary, carefully adjust the position and angle of the semi-silvered mirror). Place the appropriate filter in front of the camera lens (depending on whether you have a Hg or Na lamp) to check the interference pattern is clearly visible (see Figs. 3 and 4).

Once you have a satisfactory interference pattern think carefully about which parameters (e.g. thickness of bar, distance between knife edge and mass etc.) are required to be measured by studying the Theory Section. It is up to you to decide how accurately you need to know the various dimensions and lengths (measuring tools such as Vernier calipers and screw gauge micrometers can be obtained from the lab technician). You should always be aware of the dominant source of uncertainty and think about ways that this can be reduced.

Without further adjusting the setup, obtain images of the interference patterns associated with the applied mass ranging from 0.5 to 2.5 kg on each side of the bar. You should also obtain a calibration image by placing an object of known dimensions (e.g. a ruler) placed on top of the glass bar. This will be needed to find the distance per pixel associated with your images. If you make any changes to the setup (other than changing the mass or mass-knife edge distance) then make sure you obtain a calibration image for each setup.

7 Analysis of interference patterns

7.1 Images of Cornu fringes

An example interference pattern for this experiment is shown in Figure 4.

Question: Notice that the fringes are sets of hyperbolae. How does this fit with the description of how the bar bends in Section 5.4.

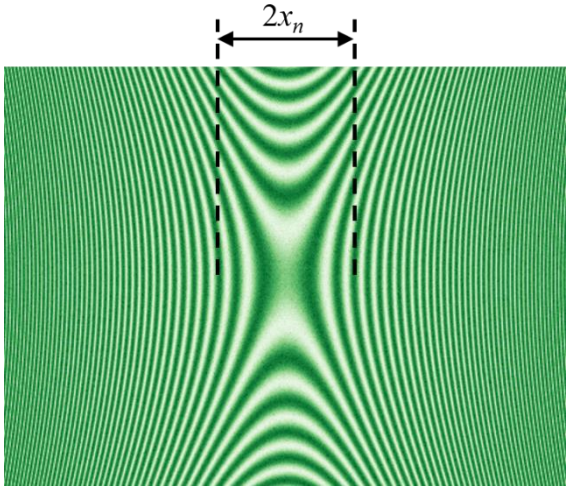


Figure 4: Example image of an interference pattern (Cornu fringes). The distance between a pair of corresponding fringes in the longitudinal direction ($2x_n$) is indicated. In this case, n=3.

Each image of an interference pattern can be used to obtain a value for longitudinal and transverse curvature, κ and κ', by measuring the horizontal and vertical distance between pairs of fringes (as illustrated in Figure 4) and then making suitable plots which will allow you to

use Equation 2 to obtain the corresponding curvatures. Some suggestions of how to measure the distances between fringes are given in Section 7.2.

While it may be tempting to use values of κ and κ' from a single image to calculate E and σ, this will not allow you to remove the effect of any residual curvature. Think about how you can combine your values using Equations 3-5 obtain a single value for E and Equation 7 to find σ such that your results are not affected by any residual curvature (in both the longitudinal and transverse directions). Ideally, you should also attempt to estimate values for the residual curvatures from your analysis.

Question: Can you modify the equations in Section 5 to account for residual curvature of the unstressed bar (i.e. non-zero values of κ and κ' when G=0.)?

Question: Equation 1 only accounts for curvature in the longitudinal direction. Write down an equation for $t(x,y)$ that includes curvature in two dimensions (i.e. in the transverse (y) direction as well).

7.2 *Measuring distances between fringes*

There are several different techniques that can be used to extract values and uncertainties of $2x_n$ from the interference patterns. You should compare the advantages and disadvantages of different methods before choosing which approach to follow. You should also consider whether repeat measurements are useful. Three approaches are outlined below. Note you do not need to implement all of these methods but you will be expected to justify your chosen approach in your assessment interview.

1) The simplest approach is by measuring distances directly from the images. This can be done on printed images using a ruler of the plots or on a computer screen (e.g. software such as Powerpoint and other graphics packages will provide the length of a line drawn over the interference pattern).

2) A more sophisticated approach is to use some form of image analysis software. **One suggested analysis tool is ImageJ** which can be downloaded from https://imagej.nih.gov/ij/. This is an easy to use application that can be run without having to install any software (e.g. it can be run within a web browser or from a USB flash drive). Full documentation on how to use ImageJ is provided on the ImageJ website, but to help you get started, once you have opened a file, use the line tool to draw across the image and then select "Plot Profile" in the "Analyze" menu to produce a plot of the intensity as a function of distance (measured in pixels). This intensity data can be exported (or simply copy and pasted) into your preferred spreadsheet and/or plotting software (e.g. Excel, OriginPro, Python etc.).

3) If you have become proficient in coding (e.g. in Python) then there are many libraries available that can be used for image processing with tools that will provide the intensity profile etc.

Whichever approach you choose, you will need to **find the calibration factor** from your calibration images (i.e. distance per pixel) to convert your measurements into actual physical distances. Note that if the camera position and settings are unchanged then each image will have the same calibration, so you only need to perform this calibration for each camera position / setup. Make sure that all uncertainties (both random and systematic) are correctly accounted for in your analysis with dominant sources of error identified and minimised where possible.

Question: Look up accepted values for Young's modulus and Poisson's ratio for glass. How do these compare with your results?

Question: Research and list other techniques for probing the elastic properties of materials. Which techniques are used for probing the microscopic mechanical properties of solid materials and which sectors of science, engineering and industry do these play an important role?

Optional extension: A bar made from Perspex can be obtained from the laboratory technician.

8 References

[1] G. F. C. Searle, *Experimental Physics* (Cambridge University Press, 1934) (Sections 55, 56, 62, 63 and footnotes to section 79). Locate a library copy.

[2] G. F. C. Searle, *Experimental Elasticity* (2nd edition) (Cambridge University Press, 1933) (Sections 27-38). Locate a library copy.

Copies of some books can be obtained from the lab technician for use during lab sessions.

Latest revision: Dr. Paul Walmsley, September 2022.

(Earlier versions: Dr. I. Duerdoth and Dr. J. Allison)